HOMEWORK 6

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1. Let (X, d) be a metric space, let $K \subset X$ be compact, and let $\mathcal{U}$ be an open cover of K. Prove that there exists a number $\delta > 0$ such that for every $x \in K$, there is some $U \in \mathcal{U}$ for which

$$B(x, \delta) \cap K \subset U.$$

Such a number δ is called a *Lebesgue number* for the cover $\mathcal{U}$ (relative to K).

Proof. Suppose for contradiction that there is no Lebesgue number. Let $a_n \in K$. Thus, for any n, $B(a_n, 1/n) \cap K \not\subseteq U$. By compactness, there exists a convergent subsequence such that $a_{n_k} \to p$. By closedness, $p \in K$. $\mathcal{U}$ covers K, so $p \in U_0$, where $U_0 \in \mathcal{U}$. Since U_0 is open, there exists an $r > 0$ such that $B(p, r) \subseteq U_0$. Now choose a k large enough so that $d(a_{n_k}, p) < r/2$ and $1/n_k < r/2$. This is true, because $1/n_k \to 0$ as $n_k \to \infty$. Now let $x \in B(a_{n_k}, 1/n_k) \cap K$. By triangle inequality, $d(x, p) \leq d(x, a_{n_k}) + d(a_{n_k}, p) < 1/n_k + r/2 < r$. Therefore, $x \in B(p, r)$. x was abitrary, so $B(a_{n_k}, 1/n_k) \cap K \subseteq U_0$. This yields a contradiction. Thus, δ exists. ∎

2. Let X be a metric space, let $K \subset X$ be a nonempty compact set, and let $p \in X \setminus K$ (K^c). Define

$$d(p, K) = \inf\{d(p, x) : x \in K\}.$$

Prove that there exists $x_0 \in K$ such that

$$d(p, K) = d(p, x_0) > 0.$$

In particular, show that a point outside a compact set has positive distance from that set.

Proof. $p \in K^c$. Since K is compact, K is closed and bounded, and its complement is open. Thus, $\exists r > 0$ such that $B(p, r) \cap K = \emptyset$. This proves that for $\forall x \in K$, $d(p, x) \geq r > 0$. Now we must show that $d(p, K)$ is attainable (infimum exists). Let $f : X \to \mathbb{R}$ be defined by $d(p, x)$, where $x \in X$. Using the reverse triangle inequality, $|d(x, p) - d(y, p)| \leq d(x, y)$. Let $d(x, y) < \epsilon$, where $\epsilon > 0$. Thus, by reverse triangle inequality, f is continuous. Now, define a sequence $x_n \in K$ such that $d(p, x_n) \to d(p, x_0)$. We can see that $d(p, x_0) \leq d(p, x_n) < d(p, x_n) + 1/n$. A subsequence converges to $x_0 \in K$, where convergence and the limit point containment are given by the compactness. Thus, $d(p, x_0) = \lim d(p, x_{n_k})$. Thus, x_0 does exist. ∎

3. Let $C_0 = [0, 1]$. Obtain C_1 by removing the open middle third $(1/3, 2/3)$ from C_0. Inductively, obtain C_n from C_{n-1} by removing the open middle third of each interval in C_{n-1}. The Cantor set is

$$C = \bigcap_{n=0}^{\infty} C_n.$$

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(a) Prove that C_n is the union of 2^n pairwise disjoint closed intervals, each of length 3^{-n}.
(b) Prove that the Cantor set C is nonempty and compact.
(c) Prove that C contains no nonempty open interval. Hence C has empty interior.

Proof. Use induction. We see $C_0 = [0, 1]$, so we have one closed interval with length 1. Now assume C_n satisfies these conditions, and deduce C_{n+1}. To obtain C_{n+1}, we take C_n and cut out the middle of each interval. This produces two disjoint intervals for each mother interval, thus giving us $2 \cdot 2^n = 2^{n+1}$, and its length would be $3^{-n}/3 = 3^{-(n+1)}$. Since the intervals in C_n are pairwise disjoint, the resulting intervals of C_{n+1} are also pairwise disjoint. Hence, C_{n+1} is the union of 2^{n+1} disjoint intervals, each of length $3^{-(n+1)}$.
Let E be the set of endpoints of all the C^n intervals. An endpoint of C_n remains an endpoint for C_{n+1}, and so on. This is because removing the middle third of the intervals never touches the endpoints. So $E \subseteq C$. E is infinite, so C is nonempty and infinite. Each C_n is a finite union of closed bounded intervals, so it is compact by Heine-Borel Theorem. Since $C \neq \emptyset$, and each C_n is closed and bounded in $[0, 1]$, so C is compact.
Now suppose for contradiction that there does exist an open interval in C. Let c be in this open interval. Thus, for $\epsilon > 0$, $B(c, \epsilon) \subseteq C$. Since $C \subseteq C_n$, $B(c, \epsilon) \subseteq C_n$. Thus, c must lie in one of those intervals. Choose $1/3^n$ such that $1/3^n < 2\epsilon$ (obtained by extracting diameter of ball). At whatever n chosen, $B(c, \epsilon) \nsubseteq C_n$. But $B(c, \epsilon) \subseteq C$, and $C \subseteq C_n$. Thus, this is a contradiction. Therefore, C contains no nonempty open interval. ■

4. Recall that a ternary expansion of $x \in [0, 1]$ has the form

$$x = \sum_{n=1}^{\infty} \frac{a_n}{3^n} = 0.a_1a_2a_3 \dots \ a_n \in \{0, 1, 2\}$$

(a) Prove that $x \in C$ if and only if x has a ternary expansion using only the digits 0 and 2. Be careful about numbers that have two ternary expansions, for example

$$\frac{1}{3} = 0.1_3 = 0.02222 \dots_3$$

(b) Use part (a) to prove that the Cantor set is uncountable.

Proof. Let $s_n = \sum_{i=1}^{n} \frac{a_i}{3^i}$ $a_i \in \{0, 2\}$. The interval containing x is $[s_n, s_n + 1/3^n]$. So $s_n \leq x \leq s_n + 1/n^3 \to 0 \leq x - s_n \leq 1/3^n$. As $n \to \infty$, $s_n \to x$ due to $1/3^n \to 0$. Hence, $x = \sum_{i=1}^{\infty} \frac{a_i}{3^n}$, which is a ternary expansion of x using only 0's and 2's.
Conversely, let $x = \sum_{i=1}^{\infty} \frac{a_i}{3^i}$ and let $s_n = \sum_{i=1}^{n} \frac{a_i}{3^i}$. Observe $x - s_n = \sum_{i=n+1}^{\infty} \frac{a_i}{3^i}$. Thus, $0 \leq x - s_n \leq \sum_{i=n+1}^{\infty} \frac{2}{3^i}$. Using the geometric series, $0 \leq x - s_n \leq 1/3^n \implies s_n \leq x \leq s_n + 1/3^n$. This interval is one of the intervals in C_n. This holds for any n, so $x \in \bigcap_{n=0}^{\infty} C_n = C$. ■

5. Let X be a metric space.

(a) Let $\{E_\alpha\}_{\alpha \in A}$ be a collection of connected subsets of X such that

$$\bigcap_{\alpha \in A} E_\alpha \neq \emptyset.$$

Prove that $\bigcup_{\alpha \in A} E_\alpha$ is connected.

(b) Let $E \subset X$ be connected and suppose

$$E \subset F \subset \overline{E}.$$

Prove that F is connected. In particular, deduce that the closure of $\overline{E}$ of a connected set is connected.

Proof. Suppose $\bigcup_{\alpha \in A} E_\alpha = A \sqcup A^c$, where A is a proper clopen set. Since $\bigcap_{\alpha \in A} E_\alpha \neq \emptyset$, let $p \in \bigcap_{\alpha \in A} E_\alpha$. From this, let $p \in A$. For each E_α, we see $p \in A \cap E_\alpha$. Since each E_α is connected and nonempty, $A \cap E_\alpha$ must be all of E_α. This is because of E_α being clopen, by inheritance principle, forces E_α to be the empty set or all of E_α. Hence $E_\alpha \subseteq A$ for every α. So $\bigcup_{\alpha \in A} E_\alpha \subseteq A$, and $A^c = \emptyset$, contradicting properness.
Prove the contrapositive. Suppose $F = A \sqcup A^c$, where A is a proper clopen subset in F. Let $K = A \cap F$ and $L = A^C \cap F$. This gives clopen subsets in F by inheritance principle. Connectedness of E forces either K or L to be the empty set. Suppose $K = \emptyset$. Let $p \in A$. Create a ball $B(p, r)$, where $r > 0$, such that $B(p, r) \cap E = \emptyset$. But $p \in F \subseteq \overline{E}$, and every ball around p intersects E. Contradiction. ■